

SIH Thermal Simulation Specification Report



Atmospheric Pressure: 67 kPa · Air Density: 0.825 kg/m³

Ambient Temperature: -8°C · Wind: 4.2 m/s

Selected Shelter Configuration:

- Shape Typology: RECTANGULAR
- Dimensions: 8m (L) × 6m (W) × 3.2m (H)
- Solar Orientation: 170° Azimuth (Due South)
- Wall Material: Insulated Panel (SIP) (k=0.035 W/mK)
- Roof Material: Insulated Roof
- Insulation Core: XPS (Extruded Polystyrene) (8 cm)

Predicted Thermal Equilibrium:

- Indoor Temperature: 31.5°C
- Heat Loss Rate: 1.98 kW
- Solar Harvest Rate: 2.18 kW
- Thermal Efficiency: 56.2%
- Comfort Band Compliance (18–24°C): MARGINAL

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